

Stress assignment without morphology in Portuguese: the qualityless vowel /0/ and the Metrically Marked Item

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Abstract:

This paper offers an Optimality-Theoretic reinterpretation and synthesis of the author's earlier Japanese works (Makino 2010, 2012) on Portuguese stress patterns, focusing exclusively on the theoretical essentials. We put forward a prosody-only account of Portuguese word stress. The analysis reduces Portuguese stress to the interaction of **a qualityless vowel /0/ carrying one mora of quantity + metrically marked items (MMIs) + general metrical theory**. This /0/ is the musical "ghost note" — **beat** (quantity) without **timbre** (quality) — which nevertheless shapes the **rhythm** (stress pattern): for example, *contador* /kont-a-dor-0/ vs. *contadora* /kont-a-dor-a/. MMIs are a class of **morpheme-sized items** that, **when the condition is satisfied, render the final syllable of the prosodic word extrametrical**.

Principles

- Trochaic binary footing from the right edge
- The rightmost foot functions as the head foot

Parameter: determination of the right edge

- **Value 1:** final syllable — unmarked default
- **Value 2:** penultimate syllable — marked (when an MMI satisfies the condition)

Metrification is computed strictly from phonological information, together with a single piece of non-categorical lexical information — namely, the **MMI condition**. No reference to syntactic categories or to morphological bracketing is made at any point in the stress assignment process. This approach therefore **eliminates morphology from stress computation proper** and unifies items such as *contador*, *contadora*, *falava*, *falávamos*, *faláveis*, *número*, *numeroso*, *numérico*, *sobre* and *após* under one minimal mechanism.

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0. Abbreviations

The abbreviations, symbols and diacritics used in this paper are as follows:

PrWd = prosodic word; **Ft** = foot; **HdFt** = head foot; **MMI** = *metrically marked morpheme*; $\sigma_{\text{final_MMI}}$ = the syllable whose nucleus is the final vowel of an MMI; **L** = light syllable; **H** = heavy syllable

/ / = underlying representation; - = morpheme boundary; /0/ = qualityless vowel with one mora of quantity; → = derivation; . = syllable boundary; Ć (e.g. ɾ, ɳ, ʂ) = consonantal syllabic nucleus that arises from the association of a consonant with the mora contributed by /0/; () = footing; ' = primary stress; , = secondary stress; [] = surface representation; # = word boundary; σ = syllable

FTBIN = foot binarity; **TROCHEE** = trochaic foot; **ALIGN-R(Ft, PrWd)** = align the right edge of each foot with the prosodic word; **ALIGN-R(Ft, $\sigma_{\text{final_MMI}}$)** = align the right edge of the foot with the syllable whose nucleus is the final vowel of an MMI; **ALIGN-R(HdFt, PrWd)** = align the right edge of the head foot with prosodic word; **PARSE(σ)** = parse every syllable into a foot; ***CLASH** = avoid a stress clash; **#WSP** = weight to stress principle

1. The qualityless vowel /0/

A **qualityless** vowel /0/ that carries **one mora** in underlying representations is assumed — e.g. *contador* /kont-a-dor-0/ (cf. *contadora* /kont-a-dor-a/); *jacaré* /zakare-0/; *trabalhar* /trabaɫ-a-r0/; *trabalharmos* /trabaɫ-a-r0-mos/; *após* /a-pɔs0/; *até* /atɛ0/. This vowel /0/ is the musical "**ghost note**" — **beat** (quantity) without **timbre** (quality) — which nevertheless shapes the **rhythm** (stress pattern). It is a purely phonological entity, serving various morphological purposes: in *contador* /kont-a-dor-0/, the /0/ functions as the thematic vowel — the stem vowel or *índice temático* (stem marker) — just as /a/ does in *contadora* /kont-a-dor-a/. On the other hand, it forms part of the infinitive or future subjunctive suffix /-r0/ in *trabalhar* /trabaɫ-a-r0/, and likewise occurs in the preposition *até* /atɛ0/, which does not display any internal morphological structure.

2. Postulation of the MMI

It is also assumed that there exists a class of *metrically marked items* (MMIs). These are **morpheme-sized forms** that are lexically specified to **render the final syllable of the PrWd extrametrical** (i.e. excluded from footing), but **only when their own final vowel serves as the nucleus of that penultimate syllable** — e.g. *-ic-* /-ik-/ in *académico* /akadɛm-ik-o/; *-vel* /-vel/ in

agradável /a-grad-a-**vel**-0/; *numer-* /numer-/ in *número* /**numɛr**-o/; -va- /-va-/ in *trabalhávamos* /trabaɫ-a-**va**-mos/; -r- /-r0-/ in *trabalhardes* /trabaɫ-a-**r0**-des/. **Whether a morpheme belongs to this class is independent of its morphosyntactic category.**

3. Unmarked metrification

Regardless of syntactic category or morphological structure, **iterative formation of trochaic binary feet proceeds from the final syllable of the PrWd towards the word-initial edge**, with the **rightmost foot** serving as the head and bearing **primary stress**:

possibilidade /pɔs-i-bil-i-dad-e/ → pɔ.si.bi.li.da.de → (,pɔ.si)(,bi.li)(da.de) → [,pu.si,βi.li'ða:.ði]
contador /kont-a-dor-0/ → kon.ta.do.ɾ → (,kon.ta)('do.ɾ) → [,kõ.te'ðo:r]
contadora /kont-a-dor-a/ → kon.ta.do.ra → (,kon.ta)('do.ra) → [,kõ.te'ðo:.rɐ]
jacaré /zakare-0/ → ʒa.ka.rɛ.0 → (,ʒa.ka)('rɛ.0) → [,ʒɐ.kɐ'rɛ:]
coração /korason-0/ → ko.ra.so.ɳ → (,ko.ra)('so.ɳ) → [,ku.rɐ'sẽũ]
trabalhamos /trabaɫ-a-mos/ → tra.ba.ɫa.mos → (,tra.ba)('ɫa.mos) → [,trɐ.βɐ'ɫɐ:.muʃ]
trabalhais /trabaɫ-a-des/ → tra.ba.ɫa.des → (,tra.ba)('ɫa.des) → [,trɐ.βɐ'ɫaiʃ]
trabalhar /trabaɫ-a-r0/ → tra.ba.ɫa.ɾ → (,tra.ba)('ɫa.ɾ) → [,trɐ.βɐ'ɫa:r]
sobre /sobre/ → so.bre → ('so.bre) → ['so:βɾi]

This suggests the following constraints in interaction: **FTBIN; TROCHEE; ALIGN-R(Ft, PrWd); ALIGN-R(HdFt, PrWd)**.

4. Marked metrification

If the **final vowel of an MMI** serves as the **nucleus of the penultimate syllable of the PrWd**, the same iterative formation of trochaic binary feet proceeds **from that penultimate syllable — rather than from the final syllable** — with the rightmost foot serving as the head foot and bearing primary stress; the final syllable is thus excluded from metrification and remains unstressed:

académico /akadem-**ik**-o/ → a.ka.dɛ.mi.**ko** → (,a.ka)('dɛ.mi)**ko** → [,ɐ.kɐ'ðɛ:.mi.ku]
agradável /a-grad-a-**vel**-0/ → a.gra.da.**ve.ɫ** → (,a.gra)('da.**ve**)**ɫ** → [,ɐ.ɣɾɐ'ða:.vɛɫ]
número /**numɛr**-o/ → **nu.mɛ.ro** → ('**nu.mɛ**)ro → ['nu:.mi.ru]
túnel /tunel-0/ → **tu.ne.ɫ** → ('**tu.ne**)**ɫ** → ['tu:.nɛɫ]
sótão /sɔtan-o/ → sɔ.**tan.o** → ('**sɔ.tan**)o → ['sɔ:ɫẽũ]

*trabalhá***vamos** /trabaʎ-a-**va**-mos/ → tra.ba.ʎa.**va**.mos → (, tra.ba)(' ʎa.**va**)mos
 → [, tre.βe ' ʎa: .vɐ.muʃ]
*trabalhá***veis** /trabaʎ-a-**va**-des/ → tra.ba.ʎa.**va**.des → (, tra.ba)(' ʎa.**va**)des → [, tre.βe ' ʎa: .vɐi̯]
*trabalhar***des** /trabaʎ-a-**r0**-des/ → tra.ba.ʎa.**r**.des → (, tra.ba)(' ʎa.**r**)des → [, tre.βe ' ʎa: .ði̯]

By contrast, if the **final vowel of an MMI** does **not function as the nucleus of the penultimate syllable of the PrWd, unmarked metrification**, as described in Section 3, emerges:

atonicidade /a-tɔn-**ik**-i-dad-e/ → a.tɔ.ni.si.da.de → (, a.tɔ)(, ni.si)(' da.de) → [, ɐ.tu.ni.si ' ða: .ði̯]
numeroso /**numɛr**-oz-o/ → **nu.mɛ**.ro.zo → (, **nu.mɛ**)(' ro.zo) → [, nu.mi ' ro: .zu]
trabalhava /trabaʎ-a-**va**/ → tra.ba.ʎa.**va** → (, tra.ba)(' ʎa.**va**) → [, tre.βe ' ʎa: .vɐ]
trabalhar /trabaʎ-a-**r0**/ → tra.ba.ʎa.**r** → (, tra.ba)(' ʎa.**r**) → [, tre.βe ' ʎa: r]

When a PrWd contains more than one MMI, metrification proceeds from the final syllable of the MMI whose final vowel function as the nucleus of the penultimate syllable of the PrWd:

numérico /**numɛr**-ik-o/ → **nu.mɛ**.ri.ko → **nu**(' mɛ.ri)ko → [nu ' mɛ: .ri.ku]

All of the above suggests the ranking: **ALIGN-R(Ft, σ_{final_MMI})** >> **ALIGN-R(Ft, PrWd)** and **ALIGN-R(HdFt, σ_{final_MMI})** >> **ALIGN-R(HdFt, PrWd)**. Note that whereas syllabification refers also to morphological information, the metrification of syllable strings refers only to phonological information.

5. Stress clash avoidance (provisional)

When the metrification domain contains an **odd** number of syllables, to avoid a stress clash between a potential **word-initial unary foot** and the following **binary foot**, metrification is assumed to proceed as follows. These generalisations are **provisional** and call for further investigation.

5.1 Five or more syllables: only the leftmost foot is ternary

Only the **leftmost** foot is **ternary**; **subsequent** feet remain **binary** trochees:

papelaria /papel-ari-a/ → pa.pe.la.ri.a → (, **pa.pe.la**)(' ri.a) → [, pɐ.pɨ.lɐ ' ða: .ði̯]
computador /konput-a-dor-0/ → kon.pu.ta.do.r̩ → (, **kon.pu.ta**)(' do.r̩) → [, kõ.pu.tɐ ' ðo: r̩]

entendimento /entend-i-ment-o/ → en.ten.di.men.to → (, **en.ten.di**)('mẽ.to) → [,ẽ.tẽ.di'ẽ:.tu]

The constraint observations and violations of the winner and the losers are as follows:

Winner # (,σσσ)...#: NO *CLASH violation; NO PARSE(σ) violation; **ONE FTBIN violation**

Loser #σ(,σσσ)...#: NO *CLASH violation; **ONE PARSE(σ) violation**; NO FTBIN violation

∴ **PARSE(σ) >> FTBIN**

Winner # (,σσσ)...#: NO *CLASH violation; NO PARSE(σ) violation; ONE FTBIN violation

Loser # (,σ)(,σσσ)...#: ONE *CLASH violation; NO PARSE(σ) violation; ONE FTBIN violation

∴ *CLASH decides (tie-breaking).

This suggests the following constraints: **PARSE(σ) >> FTBIN**; *CLASH

5.2 Initial L-H exception: initial light syllable unfooted

If a **light initial syllable** is immediately followed by a **heavy syllable**, it remains **unfooted**:

levantamento /levant-a-ment-o/ → le.van.ta.men.to → **le**(, van.ta)('men.to) → [li,vẽ.ta'mẽ:.tu]

Note that optional unary footing of the initial light syllable is not excluded, as in *levantamento* /levant-a-ment-o/ → **le**.van.ta.men.to → (, **le**)(, van.ta)('men.to) → [,li,vẽ.ta'mẽ:.tu].

The constraint observations and violations are as follows:

Winner # L (,H.σ)...#:

NO *CLASH violation; **ONE PARSE(σ) violation**; NO FTBIN violation; NO #WSP violation

Loser # (,L.H.σ)...#:

NO *CLASH violation; NO PARSE(σ) violation; ONE FTBIN violation; **ONE #WSP violation**

∴ **#WSP >> PARSE(σ)** — since PARSE(σ) >> FTBIN, as inferred in Section 5.1, the evaluation by FTBIN does not affect the evaluation by PARSE(σ).

Winner #L (,H.σ)...#:

NO *CLASH violation; **ONE PARSE(σ) violation**; NO FTBIN violation; NO #WSP violation

Loser # (,L)(,H.σ)...#:

ONE *CLASH violation; NO PARSE(σ) violation; ONE FTBIN violation; NO #WSP violation

∴ *CLASH >> PARSE(σ) — since PARSE(σ) >> FTBIN, the evaluation by FTBIN does not affect the evaluation by PARSE(σ).

This suggests the following constraint ranking: ***CLASH, #WSP** >> **PARSE(σ)** >> **FTBIN**

5.3 Three syllables: initial syllable unfooted

The **initial** syllable is **not footed**:

pessoa /peso-a/ → **pe**.so.a → **pe**('so.a) → [**pi**'so:.ɐ]
doutores /dout-or-0-s/ → **dou**.to.r0s → **dou**('to.r0s) → [**do**'to:.ɾiʃ]
pequeno /peken-o/ → **pe**.ke.no → **pe**('ke.no) → [**pi**'ke:.nu]
levar /lev-a-r0/ → **le**.va.ɾ → **le**('va.ɾ) → [**li**'va:ɾ]
até /atɛ0/ → **a**.tɛ.0 → **a**('tɛ.0) → [**ɐ**'tɛ:]
após /apɔs0/ → **a**.pɔ.s → **a**('pɔ.s) → [**ɐ**'pɔ:ʃ]

Note that optional unary footing of the initial syllable, as in *levar* /lev-a-r0/ → **le**.va.ɾ → (**le**)('va.ɾ) → [**li**'va:ɾ], is not excluded. The constraint observations and violations of the winner candidate and the loser candidates are as follows:

Winner # σ (' $\sigma\sigma$)#: **NO *CLASH violation**; **ONE PARSE(σ) violation**; NO FTBIN violation

Loser #(σ)(' $\sigma\sigma$)#: **ONE *CLASH violation**; **NO PARSE(σ) violation**; ONE FTBIN violation

∴ ***CLASH** >> **PARSE(σ)** — since **PARSE(σ)** >> **FTBIN**, the evaluation by FTBIN does not affect the evaluation by **PARSE(σ)**.

This also suggests the following constraint ranking: ***CLASH, #WSP** >> **PARSE(σ)** >> **FTBIN**

6. The constraint set (provisional)

From the discussion in Sections 3, 4 and 5, the following set of constraints and their rankings are inferred:

***CLASH, #WSP** >> **PARSE(σ)** >> **FTBIN**;
TROCHEE;
ALIGN-R(Ft, $\sigma_{\text{final_MMI}}$) >> **ALIGN-R(Ft, PrWd)**;
ALIGN-R(HdFt, PrWd) >> **ALIGN-R(HdFt, PrWd)**

7. Plausibility of the /0/ hypothesis

Plural in *-es* as /0s/ realisation

We analyse plurals such as *contadores* as /kont-a-dor-0-s/. In the singular *contador* /kont-a-dor-0/, the mora supplied by /0/ (μ_0) is associated with the final rhotic, yielding [,kõ.tɐ'ðo:r] at the surface level. In the plural, on the other hand, the /0/ in the sequence /-C0s/# surfaces as [i] through the linking of μ_0 to this inserted quality (orthographic *-es*), giving the surface [,kõ.tɐ'ðo:ri]. This analysis avoids positing an underlying /e/ in the plural — as in /kont-a-dor-e-s/ — which would have to be deleted in the singular /kont-a-dor-e/. Such a move would be incompatible with the many lexical items that retain final *-e* — e.g. *árvore, are, hectare, Zêzere* .

Northern EP corroboration: paragoge

In Northern varieties, words ending in /-r0/ or /-l0/ may surface as [-ri] ~ [-ri] or [-li] ~ [-li]. For instance, the infinitive *comer* /kõm-e-r0/ may surface as [ku'me:ri] ~ [ku'me:ri], with paragogic [i] or [i], rather than as the standard [ku'me:r]. In the standard variety, the mora supplied by /0/ is associated with [r], whereas in these Northern varieties it is associated with the epenthetic quality [i] or [i], with **the stress pattern remaining unchanged**. The "ghost note" /0/ thus occasionally acquires timbre, becoming a "full note" [i] or [i], without nevertheless altering the rhythm.

Cross-Romance corroboration: Portuguese vs. Spanish

The stress position of the future subjunctive forms *trabalharmos* and *trabalhar* can be determined solely on the basis of phonological information, by analysing the future subjunctive suffix *-r* as the MMI /-r0/ with /0/:

trabalharmos /trabaɫ-a-r0-mos/ → tra.ba.ɫa.r.mos → (,tra.ba)('ɫa.r)mos
trabalhar /trabaɫ-a-r0/ → tra.ba.ɫa.r → (,tra.ba)('ɫa.r)

This future subjunctive suffix is *-re* /-re/ with /e/ in Spanish, which is, like *-r* /-r0/ in Portuguese, an MMI and is assumed to render the final syllable of the PrWd extrametrical:

trabajáremos /trabax-a-re-mos/ → tra.ba.ja.re.mos → (,tra.ba.)('xa.re)mos
trabajare /trabax-a-re/ → tra.ba.ja.re → (,tra.ba.)('xa.re)

From this, it becomes clear that, **in stress computation, the Portuguese "ghost note" /0/ functions in exactly the same way as the Spanish "full note" /e/.**

Theoretical parsimony: economy of representation

Under the theoretical framework of this paper, the stress-pattern opposition between the quasi-minimal pair *também* and *lambem* can be easily explained by the final /0/ in *também* /tan-ben-0/, which is absent from *lambem* /lamb-e-n/:

também /tan-ben0/ → tan.be.ŋ → tan('be.ŋ) → [tẽ'bẽĩ]
lambem /lamb-e-n/ → lan.ben → ('lan.ben) → ['lẽ:.bẽĩ]

The stress-pattern difference can thus be attributed solely to the difference in their segmental sequences. Without positing /0/, by contrast, one would have to assign additionally a lexically specified accent to *também*, thereby requiring two types of information in the lexicon: segmental and lexical-accentual.

8. Orthographic corollary

When **marked metrification** emerges due to an MMI, the stressed nucleus is **typically indicated orthographically** by an acute accent or a circumflex accent, since the outcome **deviates from the default stress position** — e.g. *âmbar*, *côdea*, *número*, *pêssego*, *sótão*, *túnel*, *académico*, *agradável*, *trabalhávamos*, *trabalháveis*. This close correlation suggests that **speakers' prosodic intuitions** about **stress location** are reflected in the standard orthography, thereby providing independent support for the analysis.

9. Conclusion

Thus, Portuguese stress can be entirely reduced to **qualityless vowel /0/ with one mora of quantity + metrically marked items + general metrical theory**, eliminating the need for morphologically-driven stress assignment. The principles and parameters of primary stress assignment in Portuguese are as follows:

Principles

- Trochaic binary footing from the right edge
- The rightmost foot functions as the head foot

Parameter: determination of the right edge

- **Value 1:** final syllable — unmarked default
- **Value 2:** penultimate syllable — marked (when an MMI satisfies the condition)

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